Department of Agricultural and Applied Economics
250 Drillfield Drive
Hutcheson Hall, Suite 208
Blacksburg, Virginia 24061
P: (540) 231-6301
F: (540) 231-7417
vtagecon@vt.edu

AAEC 2434 – Foundation of Agribusiness.

COURSE INFORMATION

Credits:	3 credits
Meeting Time:	Monday and Wednesday, 2:30 p.m. -3:45 p.m.
Location:	Smyth Hall, Room 146
Instructor #1:	Dr. Walter Pangan
Office:	202-A Hutcheson Hall
Phone:	561-225-8797
Email:	walterac@vt.edu
Office Hours:	by appointment (Zoom or in person). Email is the most effective and official mode of communication for making appointments. Use "AAEC 2434" in the subject line.

Course Description

This course introduces students to the agricultural and food industries through the lens of both *new venture creation and the management of established agribusiness operations*. Students will explore fundamental economic, financial, and management tools relevant to careers in agribusiness and agriculture-related fields. Using real-world case studies and a semester-long team project, students will learn how agribusinesses are conceived, structured, and operated across the supply chain. *Key skills include interpreting and analyzing financial statements, applying budgeting and risk management tools, and developing strategies for marketing, operations, human resources, and sustainability.* By the end of the course, students will be able to connect core concepts to practical decision-making in both entrepreneurial and established agribusiness contexts.

Course Objectives

- Apply basic economic principles and business management to agribusiness firms' decision-making.
- Employ a set of planning tools that can improve the overall performance of agribusiness firms.
- Summarize key issues in organizing and managing an agribusiness such as financial performance, marketing, and human resources.
- Perform problem-solving skills through class activities involving real agribusiness scenarios.
- Prepare simple partial budgets to perform break-even analysis in agribusiness.

Recommended Textbook:

Unnaturally Delicious: How Science and Technology Are Serving Up Super Foods to Save the World by Jayson Lusk

Requirements and Grading:
The course requirements include class attendance, participation, assignments, group project, and group presentations. The weights in the final grade are assigned as follows:

Student Evaluations	**Percentage**
1. Participation (Individual)	15%
2. Assignments/Quizzes (Individual, In person, In class)	40%
3. Business Plan (Team)	
a. Proposal	5%
b. Mid-semester check point	5%
c. Final	10%
4. Presentations (Team)	
a. Proposal presentation	5%
b. Checkpoint presentation	10%
c. Final presentation	10%

Letter Grade Criteria

A 93.0-100	B+ 87.0-89.9	C+ 77.0-79.9	D+ 67.0-69.9	F <60
A– 90.0-92.9	B 83.0-86.9	C 73.0-76.9	D 63.0-66.9	
	B– 80.0-82.9	C– 70.0-72.9	D– 60.0-62.9	

Expectations:
You can expect me to:
- Challenge you to critical thinking and to understand the material.
- Engage you in the lecture and other activities to facilitate learning.
- Be available for office hours and return emails in a timely fashion, within 48 hours on business days and within 72 hours on weekends. Please use **"AAEC2434"** in your email subject line to ensure a timely response.

I expect you to:
- Attend class on time, pay attention, and be engaged.
- Be responsible for your learning, treat assignments seriously, submit them on time, and meet all important deadlines.
- Seek help from me whenever needed.

Course Policies

Late Submission and Grading
Late homework submissions will incur a penalty of 20 percentage points (out of a total of 100) per calendar day past the deadline. Assignments submitted late will not be accepted without documented evidence of force majeure. In such cases, students must email the instructor with written documentation supporting the request.

Extra Credit Policy:

Individual requests for extra credit opportunities will be denied.

Attendance Policy:
You are expected to attend classes regularly and actively participate in class discussions. You are responsible for all the announcements made in class whether you are present or not.

Civility:
Please be respectful to your professor and your fellow classmates.

Come to class on time.
Do not leave midway in the class. If you must leave early for a good reason, please sit near the door.
No text-messaging or playing with electronic devices during lectures.
No extended conversations with fellow classmates during class.
Remain in your seat throughout lecture unless otherwise instructed.

Changes in the syllabus:
The instructor reserves the right to make changes to the syllabus if desirable or necessary and will inform you via email and in class.

University Policies

Academic Integrity:
The Undergraduate Honor Code pledge that each member of the university community agrees to abide by states:

"As a Hokie, I will conduct myself with honor and integrity at all times. I will not lie, cheat, or steal, nor will I accept the actions of those who do."

Students enrolled in this course are responsible for abiding by the Honor Code. A student who has doubts about how the Honor Code applies to any assignment is responsible for obtaining specific guidance from the course instructor before submitting the assignment for evaluation. Ignorance of the rules does not exclude any member of the University community from the requirements and expectations of the Honor Code.

- **Cheating:** Cheating includes the intentional use of unauthorized materials, information, notes, study aids or other devices or materials in any academic exercise, or attempts thereof.
- **Plagiarism:** Plagiarism includes the copying of the language, structure, programming, computer code, ideas, and/or thoughts of another and passing off the same as one's own original work, or attempts thereof.
- **Falsification:** Falsification includes the statement of any untruth, either verbally or in writing, with respect to any element of one's academic work, or attempts thereof.
- **Fabrication:** Fabrication includes making up data and results, and recording or reporting them, or submitting fabricated documents, or attempts thereof.
- **Multiple Submission:** Multiple submission involves the submission for credit – without authorization from the instructor receiving the work – of substantial portions of any work (including oral reports) previously submitted for credit at any academic institution of attempts thereof.

- **Complicity:** Complicity includes intentionally helping another to engage in an act of academic misconduct, or attempts thereof.
- **Violation of University, College, Departmental, Program, Course, or Faculty Rules:** The violation of any University, College, Departmental, Program, Course, or Faculty Rules relating to academic matters that may lead to an unfair academic advantage by the student violating the rule(s).

Use of generative AI tools:
While AI tools such as ChatGPT and other AI-powered writing or research assistants can be useful, their use may impede the learning process in this course. Therefore, **the use of AI tools is prohibited unless explicitly permitted by the professor**. If allowed, all AI tool usage must be properly cited and acknowledged in your submissions. Non-compliance with these guidelines will result in a grade penalty or further disciplinary actions.

ADA Accommodation:
Virginia Tech welcomes students with disabilities into the University's educational programs. The University promotes efforts to provide equal access and a culture of inclusion without altering the essential elements of coursework. If you anticipate or experience academic barriers that may be due to disability, including but not limited to ADHD, chronic or temporary medical conditions, deaf or hard of hearing, learning disability, mental health, or vision impairment, please contact the Services for Students with Disabilities (SSD) office (540-231-3788, ssd@vt.edu, or visit ssd.vt.edu). If you have an SSD accommodation letter, please meet with me privately during office hours as early in the semester as possible to deliver your letter and discuss your accommodations. You must give me reasonable notice to implement your accommodation, which is generally 5 business days and 10 business days for final exams.

Student Well-Being:
Supporting the mental health and well-being of students in my class is of high priority to me and Virginia Tech. If you are feeling overwhelmed academically, having trouble functioning, or are worried about a friend, please reach out to any of the following offices:

- Cook Counseling: 540-231-6557 (24/7), ucc.vt.edu
- Dean of Students Office: 540-231-3787; 540-231-6411 (after hours), dos.vt.edu
- Hokie Wellness: hokiewellness.vt.edu

See a full listing of campus resources on well-being.vt.edu.

Course Outline (Tentative):

Module 1 – Understanding and Designing Agribusiness Models

- Introduction to agricultural and food industries
- A solid agribusiness model
- A business plan
- Current issues in agricultural and food industries

Module 2 – Building the Numbers (Financial Plan & KPIs)

- Introduction to financial statements
- Partial budgets
- Break-even analysis (financial focus)
- Key performance indicators

Module 3 – Economic Decision-Making in Agribusiness

- Profit maximization and breakeven conditions
- Review of supply and demand in agricultural markets
- Costs of production (fixed, variable, total, marginal)
- Market structures in the food and ag sector (perfect competition, monopolies, oligopolies)
- How economic principles guide pricing, production, and investment decisions in agribusiness

Module 4 – Quantifying the Market & Connecting with Customers

- Customer segments & needs
- Total Addressable Market
- Competitor analysis & positioning
- Quantifying your market (Serviceable and Obtainable markets)
- Sales forecasts

Module 5 – Pricing Strategies in Food and Agricultural Business

- Fundamentals of price formation in agricultural and food markets
- Cash market pricing and forward contracting
- Futures markets and hedging price risk
- Spot, forward, and formula pricing in the meat and livestock sector
- Pricing strategies for differentiated and branded food products (cost-plus, value-based)
- Linking pricing decisions to marketing, sales growth, and risk management
- Integrating pricing into the business plan and financial model (revenues, margins, and capacity)

Module 6 – Organizing the Team and the firm (Human Resources & Organizational Plan)

- Human resource decision; Team structure & culture
- The legal structure (LLC, C-corp, Sole Prioprietorship, Coop etc)

Module 7 – Planning for Risks (Risk Management & Contingency Planning)

- Risk management decision
- Mitigation strategies

Module 8 – Growing Responsibly (Sustainability & Social Responsibility)

- Environmental sustainability decision

Social impact in agribusiness